

## MATHEMATICS AND STATISTICS

TIME: 3 HOURS

[MAX. MARKS: 80]

**Note:** (i) All questions are compulsory.

(ii) Figures to the right indicate full marks.

(iii) Graph of L.P.P should be drawn on graph paper only.

(iv) Use of logarithmic table is allowed.

(v) Answers to the questions of Section – I and Section –II should be written in the only one answer book.

(vi) Answer to every new question must be written on a new page.

### Section –I

#### Question 1

[12]

(A) Select and write the appropriate answer from the given alternatives in each of the following sub-equation: [6]

(i) If  $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ , then  $A^6 = \dots\dots\dots$

- (a) 6A      (b) 12A      (c) 16A      (d) 32A

#### Solution 1:

(A)(i) Given that,

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 2I$$

$$A^6 = (2I)^6$$

$$= (2)^5 \times 2(I)^6$$

$$= (2)^5 \times 2I$$

$$= 2^5 \times A$$

$$= 32A$$

Hence, the correct option is (d).

**(ii) The principal solution of  $\cos^{-1}\left(-\frac{1}{2}\right)$  is:** (2)

- (a)  $\frac{\pi}{3}$       (b)  $\frac{\pi}{6}$       (c)  $\frac{2\pi}{3}$       (d)  $\frac{3\pi}{2}$

Solution (A) (ii):

Given that,

$$\cos^{-1}\left(-\frac{1}{2}\right) = \cos^{-1}\left[\cos\left(\pi - \frac{\pi}{3}\right)\right] = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

Hence, the correct option is (c).

**(iii) If an equation  $hxy + gx + fy + c = 0$  represents a pair of lines, then.....** (2)

- (a)  $fg = ch$       (b)  $gh = cf$       (c)  $fh = cg$       (d)  $hf = -cg$

Solution (A) (iii):

The given equation  $hxy + gx + fy + c = 0$  will represent pair of straight line,

Then,

$$\begin{vmatrix} 2 & \frac{h}{2} & \frac{-g}{2} \\ \frac{h}{2} & 0 & \frac{-f}{2} \\ \frac{-g}{2} & \frac{-f}{2} & c \end{vmatrix} = 0$$

Expand along  $R_1$

We get,

$$0 \cdot \left[ 0 - \frac{f^2}{2} \right] - \frac{h}{2} \left[ \frac{hc}{2} - \frac{gf}{4} \right] - \frac{g}{2} \left[ \frac{-fh}{4} - 0 \right] = 0$$

$$0 - \frac{h}{2} \left[ \frac{hc}{2} - \frac{gf}{4} \right] - \frac{g}{2} \left[ \frac{-fh}{4} - 0 \right] = 0$$

$$-\frac{h}{2} \left[ \frac{hc}{2} - \frac{gf}{4} \right] = \frac{g}{2} \left[ \frac{-fh}{4} \right]$$

$$-\frac{h^2c}{4} + \frac{ghf}{8} = \frac{-gfh}{8}$$

$$-\frac{h^2c}{4} = \frac{-gfh}{8} - \frac{gfh}{8}$$

$$-\frac{h^2c}{4} = \frac{-gfh}{4}$$

$$hc = gf$$

$$fg = ch$$

Hence, the correct option is (a).

**(B) Attempt any THREE of the following:**

**[6]**

**(i) Write the converse and contrapositive of the statement-**

**“If two triangle are congruent , then their areas are equal.”**

**(2)**

Solution (B) (i):

Converse: 'If areas of two triangle are equal, then they are congruent.'

Contrapositive: 'If areas of two triangles are not equal, then they are not congruent.'

**(ii) Find 'k', if the sum of slopes of lines represented by equation  $x^2 + kxy - 3y^2 = 0$  is twice their product.**

**(2)**

Solution (B) (ii):

Let the lines represented by the equation.

$$x^2 + kxy - 3y^2 = 0$$

Or,

$$-\frac{x^2}{3} - \frac{k}{3}xy + y^2 = 0 \text{ be } y = m_1x \text{ and } m_2x$$

Where  $m_1$  and  $m_2$  are the slopes of the lines.

$$\therefore -\frac{x^2}{3} - \frac{k}{3}xy + y^2 = (y - m_1x)(y - m_2x)$$

or,

$$-\frac{x^2}{3} - \frac{k}{3}xy + y^2 = y^2 - (m_1 + m_2)xy + (m_1m_2)x^2$$

Comparing both sides, we get

$$(m_1 + m_2) = \frac{k}{3} \text{ and } (m_1m_2) = -\frac{1}{3}$$

So, given that

$$m_1 + m_2 = 2m_1m_2$$

$$\therefore \frac{k}{3} = 2 \times \left(-\frac{1}{3}\right)$$

$$k = -2$$

**(iii) Find the angle between the planes  $\vec{r} \cdot (2\vec{i} + \vec{j} - \vec{k}) = 3$  and  $\vec{r} \cdot (\vec{i} + 2\vec{j} + \vec{k}) = 1$**  (2)

Solution (B) (iii):

Angle between two planes is given by

$$\cos \theta = \frac{|A_1 \cdot A_2 + B_1 \cdot B_2 + C_1 \cdot C_2|}{\sqrt{A_1^2 + B_1^2 + C_1^2} \cdot \sqrt{A_2^2 + B_2^2 + C_2^2}}$$

$$A_1 = 2, B_1 = 1, C_1 = -1$$

$$A_2 = 1, B_2 = 2, C_2 = 1$$

$$\cos \theta = \frac{|(2 \times 1) + (1 \times 2) + (-1 \times 1)|}{\sqrt{(2)^2 + (1)^2 + (-1)^2} \sqrt{(1)^2 + (2)^2 + (1)^2}}$$

$$\cos \theta = \frac{3}{6} = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3}$$

$\therefore$  Angle between the planes is  $\frac{\pi}{3}$ .

(iv) The Cartesian equation of the line are  $3x - 1 = 6y + 2 = 1 - z$ . Find the vector equation of line. (2)

**Solution (B) (iv):**

Let us consider the given of the line are:

$$3x - 1 = 6y + 2 = 1 - z$$

$$3\left(x - \frac{1}{3}\right) = 6\left(y + \frac{2}{6}\right) = (z - 1)$$

$$\Rightarrow \frac{\left(x - \frac{1}{3}\right)}{\frac{1}{3}} = \frac{\left(y + \frac{1}{3}\right)}{\frac{1}{6}} = \frac{(z - 1)}{-1} \text{-----(1)}$$

Now the general equation of the line:

$$\frac{x - a}{l} = \frac{y - b}{m} = \frac{z - c}{n} \text{-----(2)}$$

Where  $l, m$  and  $n$  are the direction ratios of the line and the point  $(a, b, c)$  lies on the line.

Compare the equation (1), with the general equation (2)

We have,

$$l = \frac{1}{3}, m = \frac{1}{6} \text{ and } n = -1$$

Also,

$$a = \frac{1}{3}, b = -\frac{1}{3} \text{ and } c = 1$$

This shows that the given line passes through

$$\left(\frac{1}{3}, -\frac{1}{3}, 1\right)$$

$$\text{Hence, } \vec{a} = \frac{1}{3}\hat{i} - \frac{1}{3}\hat{j} + \hat{k} \text{ and } \vec{b} = \frac{1}{3}\hat{i} + \frac{1}{6}\hat{j} - \hat{k}$$

The vector equation is given by:

$$r = a + \lambda b$$

$$r = \left( \frac{1}{3} \hat{i} - \frac{1}{3} \hat{j} + \hat{k} \right) + \lambda \left( \frac{1}{3} \hat{i} + \frac{1}{6} \hat{j} - \hat{k} \right)$$

(v) If  $\vec{a} = \vec{i} + 2\vec{j}$ ,  $\vec{b} = -2\vec{i} + \vec{j}$ ,  $\vec{c} = 4\vec{i} + 3\vec{j}$  find x and y such that  $\vec{c} = x\vec{a} + y\vec{b}$  (2)

**Solution (B) (v):** Given that,

$$\vec{c} = x\vec{a} + y\vec{b}$$

$$\text{Or, } 4\vec{i} + 3\vec{j} = x(\vec{i} + 2\vec{j}) + y(-2\vec{i} + \vec{j})$$

$$\text{Or, } 4\vec{i} + 3\vec{j} = (x - 2y)\vec{i} + (2x + y)\vec{j}$$

Comparing both the sides, we get

$$x - 2y = 4$$

And,

$$2x + y = 3$$

Solving the equations, we get

$$x = 2 \text{ and } y = -1$$

**Question 2** [14]

**(A) Attempt any TWO of the following** [6]

**(i) If A, B, C, D are (1, 1, 1), (2, 1, 3), (3, 2, 2), (3, 3, 4) respectively, then find the volume of the parallelepiped with AB, AC and AD as the concurrent edges. (3)**

**Solution (A) (i):**

Given that,

A, B, C and D are (1, 1, 1), (2, 1, 3), (3, 2, 2) and (3, 3, 4) respectively. We need to find the volume of the parallelepiped with AB, AC, and AD as the concurrent edges.

The volume of the parallelepiped whose edges are  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  is  $[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c})$

$$\overrightarrow{AB} = (2-1)\hat{i} + (1-1)\hat{j} + (3-1)\hat{k}$$

$$= \hat{i} + 2\hat{k}$$

$$\overrightarrow{AC} = (3-1)\hat{i} + (2-1)\hat{j} + (2-1)\hat{k}$$

$$= 2\hat{i} + \hat{j} + \hat{k}$$

$$\overrightarrow{AD} = (3-1)\hat{i} + (3-1)\hat{j} + (4-1)\hat{k}$$

$$= 2\hat{i} + 2\hat{j} + 3\hat{k}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 2 & 2 & 3 \end{vmatrix}$$

$$= 1(3-2) - 0 + 2(4-2)$$

$$= 1 + 4$$

$$= 5 \text{ cubic units}$$

**(ii) Discuss the statement pattern, using truth table:**

**(3)**

$$\sim (\sim p \wedge \sim q) \vee q$$

**Solution (A) (ii):**

We have,

| P | q | $\sim p$ | $\sim q$ | $(\sim p \wedge \sim q)$ | $\sim (\sim p \wedge \sim q)$ | $\sim (\sim p \wedge \sim q) \vee q$ |
|---|---|----------|----------|--------------------------|-------------------------------|--------------------------------------|
| 0 | 0 | 1        | 1        | 1                        | 0                             | 0                                    |
| 0 | 1 | 1        | 0        | 0                        | 1                             | 1                                    |
| 1 | 0 | 0        | 1        | 0                        | 1                             | 1                                    |
| 1 | 1 | 0        | 0        | 0                        | 1                             | 1                                    |

**(iii) If point C( $\vec{c}$ ) divides the segment joining the point A( $\vec{a}$ ) and B( $\vec{b}$ ) internally in the**

**ratio m:n, then prove that  $\vec{c} = \frac{m\vec{b} + n\vec{a}}{m+n}$**  **(3)**

**Solution (A) (iii):**

Let O be the origin,

Then,

$$\overrightarrow{OA} = \vec{a} \text{ and } \overrightarrow{OB} = \vec{b}$$

Let P be a point on  $\overrightarrow{AB}$

Such that,

$$\begin{aligned}\frac{\overrightarrow{AP}}{\overrightarrow{PB}} &= \frac{m}{n} \\ \Rightarrow n(\overrightarrow{AP}) &= m(\overrightarrow{PB}) \\ \Rightarrow n(\overrightarrow{OP} - \overrightarrow{OA}) &= m(\overrightarrow{OB} - \overrightarrow{OP}) \\ \Rightarrow (m+n)\overrightarrow{OP} &= m\overrightarrow{OB} + n\overrightarrow{OA} \\ \Rightarrow (m+n)\overrightarrow{OP} &= m\vec{b} + n\vec{a} \\ \overrightarrow{OP} = \vec{c} &= \frac{m\vec{b} + n\vec{a}}{m+n}\end{aligned}$$

Hence, proved.

**(B) Attempt any TWO of the following**

**[8]**

**(i) Find the direction cosines of the line perpendicular to the lines whose direction ratios are  $-2, 1, 1$  and  $-3, -4, 1$ .**

**(4)**

**Solution (B) (i):**

Let the direction cosines of the required line be  $l, m, n$

Thus,

$$l(-2) + m(1) + n(-1) = 0$$

$$-2l + m - n = 0$$

and,

$$l(-3) + m(-4) + n(-1) = 0$$

$$-3l - 4m + n = 0$$

By cross multiplication between (i) and (ii),

$$\frac{l}{1-4} = \frac{m}{3+2} = \frac{n}{8+3}$$

$$\frac{l}{-3} = \frac{m}{5} = \frac{n}{11}$$

Let us find,

$$\begin{aligned}\sqrt{l^2 + m^2 + n^2} &= \sqrt{(-3)^2 + (5)^2 + (11)^2} \\ &= \sqrt{9 + 25 + 121} \\ &= \sqrt{155}\end{aligned}$$

Thus, the direction ratios of the line are  $-3, 5, 11$

The direction cosines are:

$$\begin{aligned}l &= \frac{-3}{\sqrt{155}} \\ m &= \frac{5}{\sqrt{155}} \\ n &= \frac{11}{\sqrt{155}}\end{aligned}$$

**(ii) In any  $\triangle ABC$  if  $a^2, b^2, c^2$  in arithmetic progression, then prove that  $\cot A, \cot B, \cot C$  are in arithmetic progression. (4)**

**Solution (B) (ii):**

Given that,  $a^2, b^2, c^2$  are in arithmetic progression. We need to prove that  $\cot A, \cot B$  and  $\cot C$  are in A.P

$a^2, b^2, c^2$  are in A.P.

$\Rightarrow -2a^2, -2b^2, -2c^2$  are in A.P.

$\Rightarrow (a^2 + b^2 + c^2) - 2a^2, (a^2 + b^2 + c^2) - 2b^2, (a^2 + b^2 + c^2) - c^2$  are in A.P.

$\Rightarrow (b^2 + c^2 - a^2), (c^2 + a^2 - b^2), (a^2 + b^2 - c^2)$  are in A.P.

$\Rightarrow \frac{(b^2 + c^2 - a^2)}{2abc}, \frac{(c^2 + a^2 - b^2)}{2abc}, \frac{(a^2 + b^2 - c^2)}{2abc}$  are in A.P.

$\Rightarrow \frac{1}{a} \left( \frac{b^2 + c^2 - a^2}{2bc} \right), \frac{1}{b} \left( \frac{c^2 + a^2 - b^2}{2ac} \right), \frac{1}{c} \left( \frac{a^2 + b^2 - c^2}{2ab} \right)$  are in A.P.

$\Rightarrow \frac{1}{a} \cos A, \frac{1}{b} \cos B, \frac{1}{c} \cos C$  are in A.P.

$$\Rightarrow \frac{K}{a} \cos A, \frac{K}{b} \cos B, \frac{K}{c} \cos C \text{ are in A.P.}$$

$$\Rightarrow \frac{\cos A}{\sin A}, \frac{\cos B}{\sin B}, \frac{\cos C}{\sin C} \text{ are in A.P.}$$

$$\Rightarrow \cos A, \cos B, \cos C \text{ are in A.P}$$

**(iii) The sum of three numbers is 6. when second number is subtracted from thrice the sum of first and third number, we get number 10. Four times the third number is subtracted from five times the sum of first and second number, the result is 3. Using above information. Find these three numbers by matrix method. (4)**

**Solution (B) (iii):**

Let the numbers be  $x, y$  and  $z$

According to the question,

$$x + y + z = 6$$

$$\text{or, } 3(x + z) - y = 10$$

$$3x - y + 3z = 10$$

Also,

$$5(x + y) - 4z = 3$$

$$5x + 5y - 4z = 3$$

Thus,

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 3 \\ 5 & 5 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 3 \end{bmatrix}$$

Applying  $R_3 \rightarrow R_3 - 5R_1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & -1 & 3 \\ 0 & 0 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ -27 \end{bmatrix}$$

Applying  $R_2 \rightarrow R_2 - 3R_1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -4 & 0 \\ 0 & 0 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ -27 \end{bmatrix}$$

Thus,

$$x + y + z = 6$$

$$-4y = -8$$

$$y = 2$$

and,

$$-9z = -27$$

$$z = 3$$

Thus,

$$x = 6 - (2 + 3)$$

$$= 6 - 5$$

$$= 1$$

Hence,  $x = 1, y = 2$  and  $z = 3$ .

### Question 3

[14]

(A) Attempt any TWO of the following

[6]

(i) If  $\theta$  is the measure of the acute angle between the lines represented by the equation

$$ax^2 + 2hxy + by^2 = 0, \text{ then prove that } \tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| \text{ where } a + b \neq 0. \quad (3)$$

**Solution 3 (A) (i):**

Let  $m_1$  and  $m_2$  be the slope of the line represented by  $ax^2 + 2hxy + by^2 = 0$

$$\therefore m_1 + m_2 = \frac{-2h}{b}, m_1 m_2 = \frac{a}{b} \text{ where } b \neq 0$$

Consider,

$$\begin{aligned}
 (m_1 - m_2)^2 &= (m_1 + m_2)^2 - 4m_1m_2 \\
 &= \left(\frac{-2h}{b}\right)^2 - 4 \times \frac{a}{b} \\
 &= \frac{4h^2}{b^2} - \frac{4a}{b} \\
 &= \frac{4h^2 - 4ab}{b^2} \\
 &= \frac{4}{b^2}(h^2 - ab) \\
 \therefore m_1 - m_2 &= \pm \frac{2}{b}\sqrt{h^2 - ab}
 \end{aligned}$$

Let  $\theta$  be the acute angle between the lines.

$$\begin{aligned}
 \therefore \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 \cdot m_2} \right| \\
 &= \left| \frac{\pm \frac{2}{b}\sqrt{h^2 - ab}}{1 + \frac{a}{b}} \right| \\
 \therefore \tan \theta &= \left| \frac{2 \cdot \sqrt{h^2 - ab}}{a + b} \right| \\
 \therefore \theta &= \tan^{-1} \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| \text{ where } a+b \neq 0
 \end{aligned}$$

If two lines are coincident, then  $\theta = 0^\circ$ ,  $\tan \theta = \tan 0^\circ = 0$

$$\begin{aligned}
 \therefore \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| &= 0 \\
 \therefore h^2 - ab &= 0
 \end{aligned}$$

$\therefore$  The condition for coincident lines is  $h^2 - ab = 0$ .

**(ii) If the lines  $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$  and  $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{4}$  intersect each other, then find the value of 'k'.** (3)

**Solution (A) (ii):**

The given lines are:  $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$  and  $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{4}$

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} = \lambda \text{-----(1)}$$

$\therefore P(2\lambda+1, 3\lambda-1, 4\lambda+1)$  is any point on (i)

Also, let

$$\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{4} = \mu \text{-----(2)}$$

$\therefore Q(\mu+3, 2\mu+k, 4\mu)$  is any point on (ii)

The two lines will intersect if

$$2\lambda+1 = \mu+3$$

$$3\lambda-1 = 2\mu+k$$

$$4\lambda+1 = 4\mu \text{-----(iii)}$$

$$\text{or, } \mu = 2\lambda-2 \text{-----(iv)}$$

$$\text{or, } 3\lambda-2\mu = k+1 \text{-----(v)}$$

Now, substituting (iv) in (iii),

$$4\lambda+1 = 4(2\lambda-2)$$

$$\text{or, } 4\lambda+1 = 8\lambda-8$$

$$\text{or, } 4\lambda = 9$$

Substituting the value of  $\lambda$  in equation (v), we get

$$\mu = 2\left(\frac{9}{4}\right) - 2$$

$$\text{or, } \mu = \frac{9}{2} - 2$$

$$\text{or, } \mu = \frac{5}{2}$$

Substituting the value of  $\lambda$  and  $\mu$  in equation (v), we get

$$3\lambda - 2\mu = k + 1$$

$$\Rightarrow 3\left(\frac{9}{4}\right) - 2\left(\frac{5}{2}\right) = k + 1$$

$$\Rightarrow \frac{27}{4} - 5 = k + 1$$

$$\Rightarrow \frac{27}{4} - 6 = k$$

Thus,

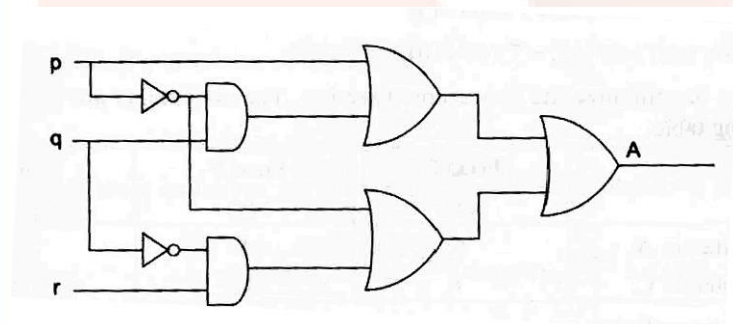
$$k = \frac{3}{4}$$

**(iii) Construct the switching circuit for the following statement:**

$$[p \vee (\sim p \wedge q)] \vee [(\sim q \wedge r) \vee \sim p] \quad (3)$$

**Solution (A) (iii):**

$$\text{Let } [p \vee (\sim p \wedge q)] \vee [(\sim q \wedge r) \vee \sim p] = A$$



**(B) Attempt any TWO of the following:**

**[8]**

$$\text{(i) Find the general solution of: } \cos x - \sin x = 1 \quad (4)$$

**Solution (A) (i):**

Given that,

$$\cos x - \sin x = 1$$

Now, dividing both sides by  $\frac{1}{\sqrt{2}}$

$$\therefore \frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x = \frac{1}{\sqrt{2}}$$

$$\cos\left(x + \frac{\pi}{4}\right) = \cos \frac{\pi}{4}$$

$$\Rightarrow x + \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4} \quad \text{where } n \in I$$

$$\Rightarrow x = 2n\pi \pm \frac{\pi}{4} - \frac{\pi}{4} \quad \text{where } n \in I$$

$$\Rightarrow x = 2n\pi, 2n\pi - \frac{\pi}{4} - \frac{\pi}{4} \quad \text{where } n \in I$$

$$\Rightarrow x = 2n\pi, 2n\pi - \frac{2\pi}{4} \quad \text{where } n \in I$$

$$\Rightarrow x = 2n\pi, 2n\pi - \frac{\pi}{2} \quad \text{where } n \in I$$

$$\Rightarrow x = 2n\pi, \frac{\pi}{2}(4n-1) \quad \text{where } n \in I$$

Hence, the general solution is  $x = 2n\pi, \frac{\pi}{2}(4n-1)$  where  $n \in I$

**(ii) Find the equations of the planes parallel to the plane  $x - 2y + 2z - 4 = 0$ , which are at a unit distance from the point  $(1, 2, 3)$ .** (4)

**Solution (A) (ii):**

Any plane II to the plane  $x - 2y + 2z - 4 = 0$  can be written as:

$$x - 2y + 2z - \lambda = 0 \text{----- (i)}$$

Where  $\lambda$  is a scalar quantity

Now, distance of the plane from the point  $(1, 2, 3)$

$$= \pm \frac{1(1) - 2(2) + 2(3) + \lambda}{\sqrt{1^2 + (-2)^2 + 2^2}}$$

$$= \pm \frac{1 - 4 + 6 + \lambda}{\sqrt{1 + 4 + 4}}$$

$$= \pm \frac{3 + \lambda}{\sqrt{9}} = \pm \frac{3 + \lambda}{3}$$

But,

$$\pm \frac{3+\lambda}{3} = 1$$

$$\Rightarrow 3+\lambda = \pm 3$$

$$\lambda = 0, -6$$

Substituting the values of  $\lambda$  in (i), we have

$$x - 2y + 2z - (-6) = 0$$

$$\text{or, } x - 2y + 2z + 6 = 0$$

**(iii) A diet of a sick person must contains at least 48 units of vitamin A and 64 units of vitamin B. Two foods  $F_1$  and  $F_2$  costs ₹10 per unit of food  $F_1$  contains ₹6 units of vitamin A and 7 units of vitamin B. One unit of food  $F_2$  costs 10 per unit. One unit of food  $F_1$  contains 6 units of vitamin A and 12 units of vitamin B. Find the minimum cost for the diet that consists of mixture of these two foods and also meeting the minimum nutritional requirements.** (4)

**Solution (A) (iii):**

Let  $x$  units of food  $F_1$  and  $y$  units of food  $F_2$  e included in the diet of the sick person.

Then their total cost is  $Z = (6x + 10y)$

Now, we have minimize the above cost function .The constraints are as per given the following table.

|           | Food $F_1$<br>(x) | Food $F_1$<br>(y) | Minimum<br>Requirement |
|-----------|-------------------|-------------------|------------------------|
| Vitamin A | 6                 | 8                 | 48                     |
| Vitamin B | 7                 | 12                | 64                     |

Hence, the constraints are

$$6x + 8y \geq 48$$

$$7x + 12y \geq 64$$

Also, the no. of units of food  $F_1$  and  $F_2$  cannot be negative.

$$x \geq 0 \text{ and } y \geq 0$$

Hence the mathematical formulation of given LPP is

$$\text{Minimize } Z = (6x + 10y)$$

Subject to

$$6x + 8y \geq 48$$

$$7x + 12y \geq 64$$

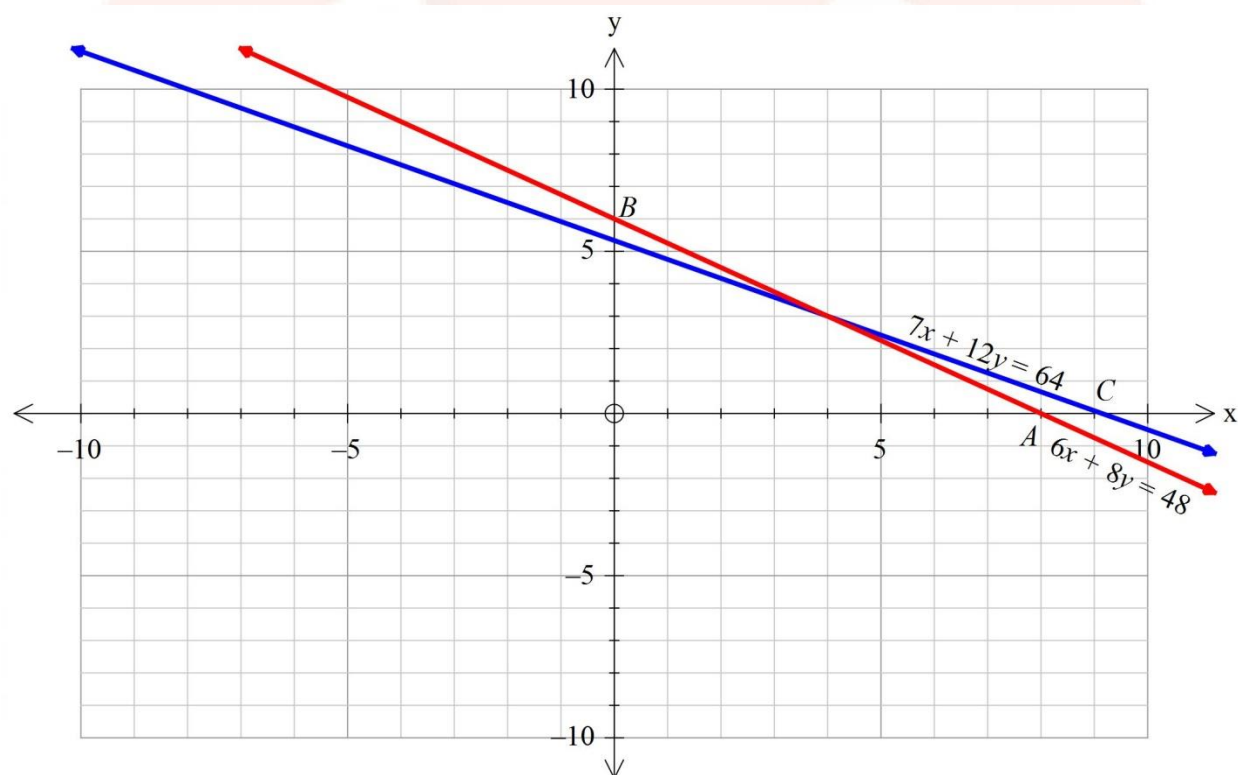
$$x \geq 0, y \geq 0$$

To solve this convert given in equations into equations we get

$$6x + 8y = 48 \Rightarrow \frac{x}{8} + \frac{y}{6} = 1$$

$$7x + 12y = 64 \Rightarrow \frac{x}{\frac{64}{7}} + \frac{y}{\frac{16}{3}} = 1$$

Plot these equation on graph paper we get feasible region shaded on graph paper



The vertices of the feasible region are  $C\left(\frac{64}{7}, 0\right)$ , P and B (0,6).

P is the point of intersection of these lines

$$6x + 8y = 48$$

$$7x + 12y = 64$$

On solving these equation we get  $P \equiv (4, 3)$ . Values of objective function at these vertices are

$$Z(C) = 6\left(\frac{64}{7}\right) + 10(0) = \frac{384}{7} = 54.85$$

$$Z(P) = 6(4) + 10(3) = 24 + 30 = 54$$

$$Z(B) = 6(0) + 10(6) = 60$$

Minimum value of  $Z$  is 54 at point  $P(4, 3)$  hence, 4 units of food  $f_1$  and 3 units of food  $f_2$  should be included in the diet of sick person to meet the minimal nutritional requirements, in order to have the minimum cost of ₹54.

## SECTION-II

**Question 4.**

**[12]**

**(A) Select and write the appropriate answer from the given alternatives in each of the following sub-questions:** **[6]**

**(i) A random variable  $X$  has the following probability distribution:** **(2)**

|         |     |     |     |     |     |     |
|---------|-----|-----|-----|-----|-----|-----|
| $X = x$ | -2  | -1  | 0   | 1   | 2   | 3   |
| $P(x)$  | 0.1 | 0.1 | 0.2 | 0.2 | 0.3 | 0.1 |

Then  $E(x) = \dots\dots\dots$

- (a) 0.8                      (b) 0.9                      (c) 0.7                      (d) 1.1

**Solution (A) (i):**

$$E(x) = \sum x.P(x)$$

|         |      |      |     |     |     |     |
|---------|------|------|-----|-----|-----|-----|
| $X = x$ | -2   | -1   | 0   | 1   | 2   | 3   |
| $P(x)$  | 0.1  | 0.1  | 0.2 | 0.2 | 0.3 | 0.1 |
| $xP(x)$ | -0.2 | -0.1 | 0   | 0.2 | 0.6 | 0.3 |
| $X = x$ | -2   | -1   | 0   | 1   | 2   | 3   |
| $P(x)$  | 0.1  | 0.1  | 0.2 | 0.2 | 0.3 | 0.1 |

$$E(x) = \sum x.P(x) = -0.2 - 0.1 + 0 + 0.2 + 0.6 + 0.3 = 0.8$$

Hence, the correct option is (a).

**(ii) If  $\int_0^a 3x^2 dx = 8$ , then the value of  $a$  is:** **(2)**

- (a) 0                      (b) -2                      (c) 2                      (d)  $\pm 2$

**Solution (A) (ii):**

Given that,

$$\int_0^a 3x^2 dx = 8$$

$$\Rightarrow 3 \int_0^a x^2 dx = 8$$

$$\Rightarrow 3 \left[ \frac{x^3}{3} \right]_0^a = 8$$

$$\Rightarrow [x^3]_0^a = 8$$

$$\Rightarrow [\alpha^3 - 0] = 2^3$$

$$\alpha = 2$$

Hence, the correct option is (c).

**(iii) The differential equation of  $y = \frac{c}{x} + c^2$  is:** (2)

(a)  $x^4 \left[ \frac{dy}{dx} \right]^2 - x \frac{dy}{dx} = y$

(b)  $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$

(c)  $x^3 \left[ \frac{dy}{dx} \right]^2 - x \frac{dy}{dx} = y$

(d)  $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$

**Solution (A) (iii):**

Given that,

$$y = \frac{c}{x} + c^2$$

Differentiating both sides of the equation w.r.t.x we get,

$$\frac{dy}{dx} = \frac{-c}{x^2}$$

$$\Rightarrow c = -x^2 \frac{dy}{dx}$$

By squaring on both sides,

$$c^2 = x^4 \left( \frac{dy}{dx} \right)^2$$

$$\therefore y = -x \frac{dy}{dx} + x^4 \left( \frac{dy}{dx} \right)^2$$

Hence, the correct option is (a).

**(B) Attempt any THREE of the following:**

**[6]**

**(i) Evaluate:**  $\int e^x \left[ \frac{\sqrt{1-x^2} \sin^{-1} x + 1}{\sqrt{1-x^2}} \right] dx$

**(2)**

**Solution (B) (i):**

Given that,

$$\int e^x \left[ \frac{\sqrt{1-x^2} \sin^{-1} x + 1}{\sqrt{1-x^2}} \right] dx$$

Let,

$$\begin{aligned} I &= \int e^x \left[ \frac{\sqrt{1-x^2} \sin^{-1} x + 1}{\sqrt{1-x^2}} \right] dx \\ &= \int e^x \left[ \sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \right] dx \end{aligned}$$

Put,

$$\begin{aligned} f(x) &= \sin^{-1} x \\ f'(x) &= \frac{1}{\sqrt{1-x^2}} \end{aligned}$$

We know that,

$$\begin{aligned} \int e^x [f(x) + f'(x)] \cdot dx &= e^x f(x) + c \\ I &= e^x \sin^{-1} x + c \end{aligned}$$

**(ii) If  $y = \sqrt{\sin x} \sqrt{\sin x + \sqrt{\sin x + \dots \dots \dots \infty}}$  then show that  $\frac{dy}{dx} = \frac{\cos x}{2y-1}$**

**(2)**

**Solution (B) (ii):** Let ,

$$y = \sqrt{\sin x \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$$

$$y = \sqrt{\sin x + y}$$

Squaring both sides , we get,

$$y^2 = \sin x + y$$

Differentiating both sides, we get

$$2y \frac{dy}{dx} = \frac{dy}{dx} + \cos x$$

$$\frac{dy}{dx} = \frac{\cos x}{(2y-1)}$$

**(iii) Evaluate:**  $\int_0^{\frac{\pi}{2}} \frac{1}{1+\cos x} dx$  (2)

**Solution (B) (iii):**

Given that,

$$\int_0^{\frac{\pi}{2}} \frac{1}{1+\cos x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} + \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2 \cos^2 \frac{x}{2}} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sec^2 \left( \frac{x}{2} \right) dx$$

Let,

$$\frac{x}{2} = u$$

$$\therefore \frac{1}{2} dx = du$$

When  $x > 0, u > 0$  and  $x > \frac{\pi}{2}, u > \frac{\pi}{4}$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sec^2 \left( \frac{x}{2} \right) dx$$

$$= \int_0^{\frac{\pi}{4}} \sec^2 u du$$

$$= [\tan u]_0^{\frac{\pi}{4}}$$

$$= \tan \frac{\pi}{4} - \tan 0 = 1$$

(iv) If  $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$ , show that  $\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$  (2)

**Solution (B) (iv):**

Given that,

$$y = e^{ax}$$

$$\Rightarrow \log y = ax$$

Differentiating both sides of equation (i) w.r.t.x, we get

$$\frac{1}{y} \cdot \frac{dy}{dx} = a$$

$$\frac{dy}{dx} = ay$$

$$\frac{dy}{dx} = ae^{ax}$$

Multiplying both sides with x, we get

$$x \frac{dy}{dx} = axe^{ax}$$

$$\Rightarrow x \frac{dy}{dx} = y \log y$$

Hence, proved.

(v) A fair coin is tossed five times. Find the probability that it shows exactly three times head. (2)

**Solution (B) (v):**

Given that,

Total no. of possible solution =  ${}^5C_3$

Total no. of solution =  $2^5 = 32$

$$\therefore P(HHH) = \frac{{}^5C_3}{32}$$

$$= \frac{10}{32} = \frac{5}{16}$$

**Question 5**

[14]

**(A) Attempt any TWO of the following:**

[6]

**(i) Integrate  $\sec^3 x$  w.r.t.x.**

(3)

**Solution 5:**

**(A) (i):**

We have,

$$\begin{aligned} \int \sec^3 x dx &= \int \sec x \sec^2 x dx \\ &= \sec x \tan x - \int \tan x \sec x \tan x dx \\ [\because \tan^2 x &= \sec^2 x - 1] \\ &= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx \\ &= \sec x \tan x + \int \sec x dx - \int \sec^3 x dx \\ \Rightarrow 2 \int \sec^3 x dx &= \sec x \tan x + \log |\sec x + \tan x| + c \\ \Rightarrow \int \sec^3 x dx &= \frac{1}{2} \sec x \tan x + \frac{1}{2} \log |\sec x + \tan x| + c \end{aligned}$$

**(ii) If  $y = (\tan^{-1} x)^2$ , show that  $(1+x^2) \frac{d^2 y}{dx^2} + 2x(1+x^2) \frac{dy}{dx} - 2 = 0$**

(3)

**Solution (A) (ii):**

Given that,

$$y = (\tan^{-1} x)^2$$

Let,

$$\tan^{-1} x = z$$

$$x = \tan z$$

Now, differentiating both sides,

$$dx = \sec^2 z dz$$

$$= (1 + \tan^2 z) dz$$

$$= (1 + x^2) dz$$

Now,

$$y = z^2$$

$$dy = 2z dz$$

$$= 2 \tan^{-1} x dz$$

$$dy = 2 \tan^{-1} x \cdot \frac{dx}{(1 + x^2)}$$

Hence,

$$\frac{dy}{dx} = \frac{2 \tan^{-1} x}{(1 + x^2)}$$

$$\Rightarrow (1 + x^2) \frac{dy}{dx} = 2 \tan^{-1} x$$

$$(1 + x^2)^2 \left( \frac{dy}{dx} \right)^2 = 4 (\tan^{-1} x)^2$$

$$(1 + x^2)^2 \left( \frac{dy}{dx} \right)^2 - 4y = 0$$

Differentiating both sides with respect to x,

$$(1 + x^2)^2 \left[ 2 \frac{dy}{dx} \right] \frac{d^2 y}{dx^2} + 2 \left( \frac{dy}{dx} \right)^2 (1 + x^2) 2x - 4 \frac{dy}{dx} = 0$$

$$\Rightarrow (1 + x^2)^2 \frac{d^2 y}{dx^2} + 2x(1 + x^2) \frac{dy}{dx} - 2 = 0$$

(iii) If  $f(x) = \left[ \tan \left( \frac{\pi}{4} + x \right) \right] \frac{1}{x}$ , for  $x \neq 0$  is continuous at  $x = 0$ , find k. (3)  
 $= k$ , for  $x = 0$

**Solution (A) (iii):**

Given that,

$$f(x) = \left[ \tan \left( \frac{\pi}{4} + x \right) \right]^{\frac{1}{x}} \quad \text{for } x \neq 0$$

$$= k \quad \text{for } x = 0$$

The function would be continuous if  $\lim_{x \rightarrow 0} f(x) = f(0)$

$$\lim_{x \rightarrow 0} \left[ \tan \left( \frac{\pi}{4} + x \right) \right]^{\frac{1}{x}} = k$$

$$\lim_{x \rightarrow 0} \left[ \frac{1 + \tan x}{1 - \tan x} \right]^{\frac{1}{x}} = k$$

$$\lim_{x \rightarrow 0} \left[ 1 + \frac{1 + \tan x}{1 - \tan x} - 1 \right]^{\frac{1}{x}} = k$$

$$\lim_{x \rightarrow 0} \left[ 1 + \frac{1 + \tan x - 1 + \tan x}{1 - \tan x} \right]^{\frac{1}{x}} = k$$

$$\lim_{x \rightarrow 0} \left[ 1 + \frac{2 \tan x}{1 - \tan x} - 1 \right]^{\frac{1}{x}} = k$$

$$\lim_{x \rightarrow 0} \left[ 1 + \frac{2 \tan x}{1 - \tan x} - 1 \right]^{\frac{1}{1 - \tan x} \times \frac{2 \tan x}{(1 - \tan x)}} = k$$

We know that,

$$\lim_{x \rightarrow 0} [1 + x]^{\frac{1}{x}} = e$$

$$\lim_{x \rightarrow 0} 2^{\frac{\tan x}{x(1-x)}} = k$$

$$e^{\lim_{x \rightarrow 0} 2 \times \frac{\tan x}{x} \times \lim_{x \rightarrow 0} \frac{1}{(1-x)}} = k \quad \left[ \because \lim_{x \rightarrow 0} \left( \frac{\tan x}{x} \right) = 1 \right]$$

$$e^{2 \times 1 \times 1} = k$$

$$\therefore k = e^2$$

**(B) Attempt any TWO of the following:**

**[8]**

**(i) Find the coordinates of the points on the curve  $y = x - \frac{4}{x}$ , where the tangents are parallel to the line  $y = 2x$ .**

**(4)**

**Solution (B) (i):**

Equation of line is given by

$$y = 2x \text{ -----(1)}$$

We know that

$$y = mx + c \text{ -----(2)}$$

Now, the comparing equation (2) with (1)

We get,

$$m = 2$$

Given,

$$y = x - \frac{4}{x}$$

Differentiating w.r.to.x,

$$\begin{aligned} \frac{dy}{dx} &= 1 - 4(-1) \cdot \frac{1}{x^2} \\ &= 1 + \frac{4}{x^2} = m \text{ -----(3)} \end{aligned}$$

Slop of the tangent,

$$m = 1 + \frac{4}{x^2}$$

$$1 + \frac{4}{x^2} = 2$$

$$x^2 + 4 = 2x^2$$

$$x \pm 2$$

$$\therefore x_1 = 2, x_2 = -2$$

$$y_1 = 4, y_2 = -4$$

Co-ordinates of the point of contact are (2, 4) and (-2, -4).

**(ii) Prove that:**  $\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + c$  (4)

**Solution (B) (ii):**

Given that,

$$\begin{aligned} & \int \sqrt{x^2 - a^2} dx \\ &= \sqrt{x^2 - a^2} (x) - \int \frac{2x}{2\sqrt{x^2 - a^2}} x dx \\ &= \sqrt{x^2 - a^2} (x) - \int \frac{x^2}{\sqrt{x^2 - a^2}} dx \\ &= \sqrt{x^2 - a^2} (x) - \int \frac{x^2 - a^2 + a^2}{\sqrt{x^2 - a^2}} dx \\ &= \sqrt{x^2 - a^2} (x) - \int \frac{x^2 - a^2}{\sqrt{x^2 - a^2}} dx + a^2 \int \frac{1}{\sqrt{x^2 - a^2}} dx \\ &= \sqrt{x^2 - a^2} (x) - \int \sqrt{x^2 - a^2} dx + a^2 \int \frac{1}{\sqrt{x^2 - a^2}} dx \\ &\Rightarrow 2 \int \sqrt{x^2 - a^2} dx = \sqrt{x^2 - a^2} (x) + a^2 \int \frac{1}{\sqrt{x^2 - a^2}} dx \\ &\Rightarrow \int \sqrt{x^2 - a^2} dx = \frac{1}{2} \sqrt{x^2 - a^2} (x) + \frac{1}{2} a^2 \log \left| x + \sqrt{x^2 - a^2} \right| + c \end{aligned}$$

Where c is constant term.

**(iii) Evaluate:**  $\int_0^\pi \frac{x \sin x}{1 + \sin x} dx$  (4)

**Solution (B) (iii):**

Let,

$$\int_0^\pi \frac{x \sin x}{1 + \sin x} dx \text{----- (i)}$$

Then,

$$I = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \sin(\pi - x)} dx$$

$$= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \sin x} dx \text{-----(ii)}$$

Adding equation (i) and (ii),

$$2I = \int_0^{\pi} \frac{\pi \sin x}{1 + \sin x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \sin x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{(1 - \sin x) \sin x}{(1 + \sin x)(1 - \sin x)} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x - \sin^2 x}{1 - \sin^2 x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x - \sin^2 x}{\cos^2 x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{\cos^2 x} - \frac{\pi}{2} \int_0^{\pi} \frac{\sin^2 x}{\cos^2 x} dx$$

$$\Rightarrow I = \frac{\pi}{2} \int_0^{\pi} \tan x \sec x dx - I = \frac{\pi}{2} \int_0^{\pi} \frac{1 - \cos^2 x}{\cos^2 x} dx$$

$$\Rightarrow I = \frac{\pi}{2} [\sec x]_0^{\pi} - \frac{\pi}{2} \int_0^{\pi} \sec^2 x dx + \frac{\pi}{2} \int_0^{\pi} dx$$

$$\Rightarrow I = \frac{\pi}{2} [\sec \pi - \sec 0] - \frac{\pi}{2} [\tan x]_0^{\pi} + \frac{\pi}{2} [x]_0^{\pi}$$

$$\Rightarrow I = \frac{\pi}{2} [-1 - 1] - \frac{\pi}{2} [\tan \pi - \tan 0] + \frac{\pi}{2} [\pi - 0]$$

$$\Rightarrow I = -\pi - \frac{\pi}{2} [0 - 0] + \frac{\pi}{2} [\pi]$$

$$\Rightarrow I = \pi \left[ \frac{\pi}{2} - 1 \right]$$

Hence,

$$\int_0^{\pi} \frac{x \sin x}{1 + \sin x} dx = \pi \left[ \frac{\pi - 2}{2} \right]$$

**Question 6.**

**[14]**

**(A) Attempt any TWO of the following:**

**[6]**

(i) Find a and b, so that the function f(x) defined by

(3)

$$f(x) = -2 \sin x, \text{ for } -\pi \leq x \leq -\frac{\pi}{2}$$

$$= \cos x, \text{ for } \frac{\pi}{2} \leq x \leq \pi$$

Is continuous on  $[-\pi, \pi]$

**Solution 6:**

**(A) (i):**

It is given that f is continuous on  $[-\pi, \pi]$

$\therefore$  f is continuous at  $x = -\frac{\pi}{2}$  and  $\frac{\pi}{2}$

Continuity at  $x = -\frac{\pi}{2}$

$$f(x) = -2 \sin x, \text{ for } -\pi \leq x \leq -\frac{\pi}{2}$$

$$\therefore f\left(-\frac{\pi}{2}\right) = -2 \sin\left(-\frac{\pi}{2}\right) = 2 \sin \frac{\pi}{2}$$

$$= 2$$

$$f(x) = a \sin x + b, \text{ for } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \text{-----(1)}$$

$$\therefore \lim_{x \rightarrow \left(-\frac{\pi}{2}\right)^+} f(x) = \lim_{x \rightarrow \left(-\frac{\pi}{2}\right)^+} (a \sin x + b)$$

$$= a \sin\left(-\frac{\pi}{2}\right) + b$$

$$= -a + b$$

so,

$$-a + b = 2 \text{-----(2)}$$

Continuity at  $x = \frac{\pi}{2}$

From (i)

$$\lim_{x \rightarrow \left(\frac{\pi}{2}\right)^+} f(x) = \lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} (a \sin x + b)$$

$$= -a + b$$

$$f(x) = \cos x \quad \text{for } \frac{\pi}{2} < x < \pi$$

$$\therefore f\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2} = 0$$

$$\therefore a + b = 0 \text{------(3)}$$

Now, Adding (2) and (3) we get,

$$2b = 2 \Rightarrow b = 1$$

and,

$$a = -1$$

Hence,

$$a = -1 \text{ and } b = 1$$

**(ii) If  $\log_{10} \left( \frac{x^3 - y^3}{x^3 + y^3} \right) = 2$ , then show that  $\frac{dy}{dx} = \frac{-99x^2}{101y^2}$ . (3)**

**Solution (A) (ii):**

Given that,

$$\log_{10} \left( \frac{x^3 - y^3}{x^3 + y^3} \right) = 2$$

$$\Rightarrow \log_{10}(x^3 - y^3) - \log_{10}(x^3 + y^3) = 2$$

Now, differentiating with respect to x, we have,

$$\begin{aligned} & \frac{1}{x^3 - y^3} \left( 3x^2 - 3y^2 \frac{dy}{dx} \right) - \frac{1}{x^3 + y^3} \left( 3x^2 + 3y^2 \frac{dy}{dx} \right) \\ \Rightarrow & \frac{3x^2}{x^3 - y^3} - \frac{3y^2}{x^3 - y^3} \frac{dy}{dx} - \frac{3x^2}{x^3 + y^3} - \frac{3y^2}{x^3 + y^3} \frac{dy}{dx} = 0 \\ & -\frac{3y^2}{x^3 - y^3} \frac{dy}{dx} - \frac{3y^2}{x^3 + y^3} \frac{dy}{dx} = -\frac{3x^2}{x^3 - y^3} + \frac{3x^2}{x^3 + y^3} \\ & -3y^2 \frac{dy}{dx} \left[ -\frac{1}{x^3 - y^3} + \frac{1}{x^3 + y^3} \right] = -3x^2 \left[ \frac{1}{x^3 - y^3} - \frac{1}{x^3 + y^3} \right] \\ & y^2 \frac{dy}{dx} \left[ -\frac{1}{x^3 - y^3} + \frac{1}{x^3 + y^3} \right] = x^2 \left[ \frac{1}{x^3 - y^3} - \frac{1}{x^3 + y^3} \right] \\ & y^2 \frac{dy}{dx} \left[ \frac{x^3 + y^3 + x^3 - y^3}{(x^3 - y^3)(x^3 + y^3)} \right] = x^2 \left[ \frac{x^3 + y^3 - x^3 + y^3}{(x^3 - y^3)(x^3 + y^3)} \right] \\ & y^2 \frac{dy}{dx} \left[ \frac{2x^3}{(x^3 - y^3)(x^3 + y^3)} \right] = x^2 \left[ \frac{2y^3}{(x^3 - y^3)(x^3 + y^3)} \right] \\ & x \frac{dy}{dx} = y \\ & \frac{dy}{dx} = \frac{y}{x} \text{-----(i)} \end{aligned}$$

Also,

$$\begin{aligned} \log_{10} \frac{x^3 - y^3}{x^3 + y^3} &= 2 \\ \frac{x^3 - y^3}{x^3 + y^3} &= 10^2 \\ \frac{x^3 - y^3}{x^3 + y^3} &= 100 \\ x^3 - y^3 &= 100(x^3 + y^3) \\ x^3 - y^3 &= 100x^3 + 100y^3 \\ -99x^3 &= 101y^3 \\ \frac{y^3}{x^3} &= -\frac{99}{101} \\ \frac{y}{x} &= \frac{-99x^2}{101y^2} \text{-----(ii)} \end{aligned}$$

Hence, proved

**(iii) Let the p.m.f (probability mass function) of random variable x be (3)**

$$P(x) = \left(\frac{4}{x}\right) \left(\frac{5}{9}\right)^x \left(\frac{4}{x}\right)^{4-x}, x = 0, 1, 2, 3, 4$$

$$= 0, \text{otherwise}$$

**Find E(x) and Var.(x)**

**Solution (A) (iii):**

Given that,

$$P(x) = \left(\frac{4}{x}\right) \left(\frac{5}{9}\right)^x \left(\frac{4}{x}\right)^{4-x}, x = 0, 1, 2, 3, 4$$

$$= 0, \text{otherwise}$$

Comparing this with  $P(x) = {}^nC_x p^x q^{n-x}$  we get,

$$n = 4, p = \frac{5}{9}, q = \frac{4}{9}$$

$$E(x) = np = 4 \left(\frac{5}{9}\right) = \frac{20}{9}$$

$$= 2.22$$

$$Var(x) = npq = 4 \times \frac{5}{9} \times \frac{4}{9}$$

$$= \frac{80}{81}$$

$$= 0.9876$$

Hence,  $E(x) = 2.22$  and  $Var(x) = 0.9876$

**(B) Attempt any TWO of the following: [8]**

**(i) Examine the maxima and minima of the function  $f(x) = 2x^3 - 21x^2 + 36x - 20$ . Also, find the maximum and minimum value of f(x). (4)**

**Solution (B) (i):**

Given that,

$$f(x) = 2x^3 - 21x^2 + 36x - 20$$

$$f'(x) = 6x^2 - 42x + 36$$

$$f''(x) = 12x - 42$$

Taking,

$$f'(x) = 0$$

$$\therefore 6x^2 - 42x + 36 = 0$$

$$\Rightarrow 6(x^2 - 7x + 6) = 0$$

$$\Rightarrow x^2 - 7x + 6 = 0$$

$$\Rightarrow (x-6)(x-1) = 0$$

$$\therefore x = 6 \text{ or } x = 1$$

now,

$$f''(x=6) = 12 \times 6 - 42 = 72 - 42 = 30 > 0$$

$\therefore f(x)$  has minima at  $x = 6$

Min. value of  $f(x)$  at  $x = 6$  is

$$f(6) = 2(6)^3 - 21(6)^2 + 36 \times 6 - 20$$

$$= 2 \times 216 - 21 \times 36 + 36 \times 6 - 20$$

$$= 432 - 756 + 216 - 20$$

$$= -128$$

$$f'(x=1) = 12 \cdot 1 - 42 = 12 - 42 = -30 < 0$$

$f(x)$  has maxima at  $x = 1$

Max. value of  $f(x)$  at  $x = 1$  is

$$f(1) = 2 \cdot 1^3 - 21 \cdot 1^2 + 36 \cdot 1 - 20$$

$$= 2 - 21 + 36 - 20$$

$$= 38 - 41$$

$$= -3$$

**(ii) Solve the differential equation  $(x^2 + y^2)dx - 2xydy = 0$**

**(4)**

**Solution (A) (ii):**

Given that,

$$(x^2 + y^2)dx - 2xydy = 0$$

$$\Rightarrow (x^2 + y^2)dx = 2xydy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + y^2}{2xy} \text{----- (i)}$$

let  $y = vx$

Thus,

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Thus,

$$v + x \frac{dv}{dx} = \frac{x^2 + (vx)^2}{2x(vx)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2 - 2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - v^2}{2v}$$

$$\Rightarrow \frac{dx}{x} = \frac{2v}{1 - v^2} dv$$

$$\Rightarrow \frac{dx}{x} - \frac{2v}{1 - v^2} dv = 0 \text{----- (ii)}$$

Integrating both sides,

$$\log x + \log(1 - v^2) = \log C$$

$$\Rightarrow \log x(1 - v^2) = \log C$$

$$\Rightarrow x(1 - v^2) = C$$

$$\Rightarrow x \left( 1 - \frac{y^2}{x^2} \right) = C$$

$$\Rightarrow x \left( \frac{x^2 - y^2}{x^2} \right) = C$$

$$\Rightarrow x^2 - y^2 = Cx$$

(iii) Given the p.d.f (probability density function) of a continuous random variable  $x$  as: (4)

$$f(x) = \frac{x^2}{3}, -1 < x < 2$$

$$= 0, \text{ otherwise}$$

Determine the c.d.f (cumulative distribution function) of  $x$  and hence find

$$P(x < 1), P(x \leq -2), P(x > 0), P(1 < x < 2)$$

**Solution (A) (iii):**

c.d.f of the continuous variable is given by

$$f(x) = \int_{-1}^x \frac{y^2}{3}$$

$$= \left[ \frac{y^3}{9} \right]_{-1}^x$$

$$\left( \frac{x^3 + 1}{9} \right), x \in R$$

Consider,

$$P(x < 1) = f(1) = \frac{(1)^3 + 1}{9} = \frac{2}{9}$$

$$P(x \leq -2) = 0$$

$$P(x > 0) = 1 - P(x \leq 0)$$

$$= 1 - f(0) = 1 - \left( \frac{1}{9} \right) = \frac{8}{9}$$

$$P(1 < x < 2) = f(2) - f(1) = 1 - \frac{2}{9} = \frac{7}{9}$$